

12.432

EE25BTECH11002 - Achat Parth Kalpesh

Question:

If any two columns of $\mathbf{p} = \begin{pmatrix} 4 & 7 & 8 \\ 3 & 1 & 5 \\ 9 & 6 & 2 \end{pmatrix}$ are interchanged, which one of the following statements regarding the value of the determinant is CORRECT?

- 1) Absolute value remains unchanged but sign will change.
- 2) Both absolute value and sign will change.
- 3) Absolute value will change but sign will not change.
- 4) Both absolute value and sign will remain unchanged.

Solution:

Let

$$\mathbf{P} = [\mathbf{p}_1 \ \mathbf{p}_2 \ \mathbf{p}_3] \quad (4.1)$$

where

$$\mathbf{p}_1 = \begin{pmatrix} 4 \\ 3 \\ 9 \end{pmatrix}, \quad \mathbf{p}_2 = \begin{pmatrix} 7 \\ 1 \\ 6 \end{pmatrix}, \quad \mathbf{p}_3 = \begin{pmatrix} 8 \\ 5 \\ 2 \end{pmatrix} \quad (4.2)$$

Then

$$\det(\mathbf{P}) = \det(\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3) \quad (4.3)$$

Determinant of $\mathbf{P}$ is

$$\det(\mathbf{P}) = 4(1 \cdot 2 - 5 \cdot 6) - 7(3 \cdot 2 - 5 \cdot 9) + 8(3 \cdot 6 - 1 \cdot 9) \quad (4.4)$$

$$= 4(2 - 30) - 7(6 - 45) + 8(18 - 9) \quad (4.5)$$

$$= 4(-28) - 7(-39) + 8(9) \quad (4.6)$$

$$= -112 + 273 + 72 \quad (4.7)$$

$$= 233 \quad (4.8)$$

Hence,

$$\det(\mathbf{P}) = 233 \quad (4.9)$$

Determinant after interchanging columns 1 and 2 is $\mathbf{P}'$

$$\mathbf{P}' = [\mathbf{p}_2 \ \mathbf{p}_1 \ \mathbf{p}_3] = \begin{pmatrix} 7 & 4 & 8 \\ 1 & 3 & 5 \\ 6 & 9 & 2 \end{pmatrix} \quad (4.10)$$

Then

$$\det(\mathbf{P}') = 7(3 \cdot 2 - 5 \cdot 9) - 4(1 \cdot 2 - 5 \cdot 6) + 8(1 \cdot 9 - 3 \cdot 6) \quad (4.11)$$

$$= 7(6 - 45) - 4(2 - 30) + 8(9 - 18) \quad (4.12)$$

$$= 7(-39) - 4(-28) + 8(-9) \quad (4.13)$$

$$= -273 + 112 - 72 \quad (4.14)$$

$$= -233 \quad (4.15)$$

Thus,

$$\det(\mathbf{P}') = -233 \quad (4.16)$$

$$\det(\mathbf{P}') = -\det(\mathbf{P}) \quad (4.17)$$

Hence, on interchanging any two columns of a matrix:

The sign of the determinant changes, but the absolute value remains the same